Emily Tymafichuk

Burnaby, British Columbia
587-377-1031
emt8@sfu.ca

Skill Set

- **Technical:** Python, C, C++, Swift, HTML/CSS, JavaScript, JSON, VS Code
- **Web Development**: Full-Stack Applications, Interactive Prototyping, Accessible and Engaging UI/UX Design, Scalable and Cross-Compatible Implementations
- **Programming**: Modular and Maintainable Code, Efficient Troubleshooting Methods, Large-Scale Data Structures, Collaborative Project Development

Academic Technical Projects

Wordlist AVL Tree — November 2024
Data Structures and Algorithms (CMPT 225), SFU

- Implemented an AVL tree-based Wordlist class to efficiently store and process 200,000+ words, ensuring O(log n) operations for insertions, deletions, and lookups, improving scalability and performance.
- Implemented file I/O processing to parse and process large datasets, accurately inserting words into the AVL tree and generating word frequency statistics.
- Optimized AVL tree duplication with a recursive deep copy mechanism and in-order traversal for efficient lexicographic word output and frequency retrieval.

Blackfoot Language App — December 2023
Introduction to Computer Science and Programming (CMPT 120), SFU

- Developed an audio-visual language learning game using Python and Pygame to support Blackfoot language preservation.
- Implemented image processing functionality, including recoloring, mirroring, and resizing, to enhance the learning experience and user engagement.
- Simulated real-world learning by randomly displaying images and playing it's corresponding audio to reinforce word-object association.

Personal Technical Projects

Somnio — November 2024

- Independently developed Somnio: An iOS app designed to provide real-world solutions by calculating users' ideal sleep and wake times based on natural sleep cycles using Swift/SwiftUI.
- Designed a user-friendly interface with a logical navigation system to enhance usability and accessibility through intuitive design choices.
- Managed development tasks by prioritizing and scheduling them strategically for efficiency and alignment with project goals.

Emily Tymafichuk

Page 2

Personal Technical Projects Cont.

StudyStreak — October 2024
SFU Surge StormHacks

- Collaborated with a team to create StudyStreak: A student-oriented platform promoting social good by incentivizing consistent self-improvement.
- Implemented key features, including a task tracker, leaderboard, and achievements, using HTML, CSS, and JavaScript to enhance user engagement.
- Led prototype development through delegation, cooperation, and task prioritization to ensure timely completion.

Get Home — September 2024
Fall Hacks Hackathon Award

- Earned “Best Use of Theme” award by designing visual components that enhanced user appeal and aligned creatively with a central theme “Faster than Light”.
- Collaborated with a team to develop Get Home: An interactive application featuring multimedia elements and a health-bar-style game play using Python and Pygame.
- Swiftly resolved technical challenges through troubleshooting methods, ensuring a polished and functional final product.

Professional Experience

Back-End Client Care Intern — July - August 2024
True Balance Medical Spa

- Navigated multiple business management platforms (including Zenoti, HubSpot, and Podium) to streamline appointment scheduling,
- Handled client inquiries professionally to promote a positive customer experience while maintaining trust and brand integrity.
- Contacted 300+ clients daily during seasonal promotions, ensuring timely and accurate communication for maximal engagement.

Education

Bachelor of Computing Science — September 2023 - Present
Simon Fraser University, Burnaby, British Columbia

- Minor in Mathematics
- GPA: 3.74